

2208A **API**

BLE wristband (2208A)
Communication Protocol Specification

Version: R0 1
Time: 20 2 3 . 0 4 . 2 6

API

Protocol description

This agreement is only valid for BLE Pedometer communication protocol interpretation.

1. BLE Profile description

Service UUID: 0xFFFD

Characteristic UUID:

TX UUID: 0xFFD6 *The mobile phone sends data to the bracelet, the length is 16 bytes*

RX UUID: 0xFFD7 *The bracelet sends data to the mobile phone, the length is 16 bytes*

BLE communication only interacts between two characteristic UUIDs, see Section 2 for the specific protocol format.

2. BLE Protocol

A). The upper computer sends a 16-byte data packet to the lower computer, and then the lower computer sends back a 16-byte data packet. All but a few commands utilize this method of communication.

B). The format of the 16-byte data packet sent by the host computer is as follows:

B1	B2 ----- B15	B16
Order	payload bytes, 14 bytes	CRC check byte

Command: value range 0x00 → 0x7F; Bit7 is always 0

The last byte is CRC. The calculation of CRC is simply to add all the previous 15 bytes, and then take the lowest 8 bits. For example, if the data packet is:

A1 A2 A3 A4..... A13 A14 A15 CRC, then

$CRC = (A1 + A2 + A3 + A4 + \dots + A13 + A14 + A15) \& 0xFF$;

C). The lower computer returns a 16-byte response packet after receiving the command, the format is as follows

B1	B2 ----- B15	B16
Order	payload bytes, 14 bytes	CRC check byte

command: Reply to a command received.

If the 0x01 command is received and the CRC check is OK, then return 0x01, if the CRC is wrong, set Bit7 to 1, and return 0x81.

The last byte is CRC. The calculation of CRC is simply to add all the previous 15 bytes, and then take the lowest 8 bits. For example, if the data packet is:

A1 A2 A3 A4..... A13 A14 A15 CRC, then

$CRC = (A1 + A2 + A3 + A4 + \dots + A13 + A14 + A15) \& 0xFF$;

3. BLE Data Format

1. Set the time

Command format: 0x01 AA BB CC DD EE FF GG 00 00 00 00 00 00 00 CRC

Function: Write the current time of the PC to the bracelet, a total of 16 bytes

Description: AA year; BB month; CC day; DD hour; EE minute; FF second. The format is BCD format, such as 12 years, AA = 0x12

CRC

AA: Maximum length of Bluetooth data packet

GG: Maximum length of Bluetooth data packet

0x83: Enable ANCS and enable ANCS debugging function.

API

After the ANCS debugging function is enabled, the debugging command 0X70 will be returned when receiving the information; the format is as follows:

70 AA BB CC S1 S2 ...

AA: Whether ANCS is turned on, BB CC sub-switch status. S1 S2. . . String, returns the received manufacturer information.

GG HH: ANCS information switch, 15 digits in total. The highest bit is used to control whether a separate information switch needs to be set. The arrangement of the detailed sub-switches is as follows. The first is the lowest bit

Incoming call (lowest digit), "Call", //Incoming call
"Wechat",

"Message", //information
"Facebook",

"WhatsApp",
"Telegra", //
"QQ",

II: Basic heart rate setting, the highest bit is 1 and the value must exceed 40

command reply:

The verification is correct and the execution OK returns: 0x03 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 CRC

Verification error or execution Fail returns: 0x83 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 CRC

4. Obtain the basic parameters of the bracelet:

Command format: 0x04 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 CRC

Function: Get the basic parameters of the bracelet

command reply:

The verification is correct and the execution OK returns: 0x04 AA BB CC DD EE FF GG HH II JJ KK LL
MM NN CRC

FF: ANCS enable switch: 0x81: enable, 0x80: disable

0x83: Enable ANCS and enable ANCS debugging function.

After turning on the ANCS debugging function, the debugging command 0X70 will be returned when receiving information; the format is as follows:

70 AA BB CC S1 S2

AA: Whether ANCS is on, BB CC sub-switch status. S1 S2. . . String, return received manufacturer information.

GG HH: ANCS information switch, 15 digits in total. The highest bit is used to control whether a separate information switch needs to be set. The arrangement of the detailed sub-switches is as follows. The front one is the lowest one

Incoming call (lowest digit), "Call", //Incoming call
"Wechat",

"Message", //Message
"Facebook",

"WhatsApp",

API

"Telegra" , //

"QQ" ,

II: Basic heart rate setting, the highest bit is 1 and the value must exceed 40

Verification error or execution Fail returns: 0x84 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 CRC

5. Set the device ID code:

Command format: 0x05 ID5 ID4 ID3 ID2 ID1 ID0 00 00 00 00 00 00 00 00 CRC

Function: PC writes the ID code into the device;

Description: The 6 Bytes ID code is arranged with the high byte first;

command reply:

The verification is correct and the execution OK returns: 0x05 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 CRC

Verification error or execution Fail returns: 0x85 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 CRC

6. Real-time **step counting, heart rate, temperature** mode

Command format: 0x09 AA BB 00 00 00 00 00 00 00 00 00 00 00 00 CRC

Function: Enable or disable real-time step counting mode

AA: 1: Start real-time step counting, 0: Turn off real-time step counting

B B: 1: Turn on the real-time temperature mode. 0: Disable real-time temperature mode.

Command reply:

The verification is correct and the execution is OK and returns: 0x09 AA BB CC DD EE FF GG HH II JJ
KK LL MM NN OO PP QQ RR SS TT UU T1 T2 VV .

Verification error or execution Fail returns: 0x89 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 CRC

Reply description:

After the startup is successful, when the bracelet data changes, it will automatically retransmit the command reply

AA BB CC DD is the total number of steps in 4 bytes, the high bit comes after

EE FF GG HH is a 4-byte calorie value, with the high bit following. (When displaying calories, please divide by 100, keep two decimal places, and the unit is KCAL)

II JJ KK LL is a 4-byte walking distance, with the high bit following. (When displaying the distance, please divide by 100, keep two decimal places, and the unit is KM)

MM NN OO PP is the movement time of 4 bytes, the high bit comes after, and the unit is minute and second.

QQ RR SS TT is the fast motion time of 4 bytes, the high bit comes after, and the unit is minute.

UU: : 1 byte, heart rate value.

T1 T2: temperature value, if it is turned on to detect the temperature in real time, it will be updated every 1 second , if it is not turned on, the historical data will be transmitted.

VV: 1 byte blood oxygen value 0~100

7. Set goals:

API

Command format:	0x0B AA BB CC DD 00 00 00 00 00 00 00 00 00 00 CRC
Function:	Write the target number of steps to the bracelet
Description:	AA BB CC DD is the target step value, 4 bytes, low byte first

command reply:

The verification is correct and the execution OK returns: 0x0B 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 CRC

Verification error or execution Fail return: 0x8B 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 CRC

8. Achieve goals:

Command format:	0x4B 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 CRC
Function:	The mobile phone reads the target step number from the device

command reply:

The verification is correct and the execution OK returns: 0x4B AA BB CC DD 00 00 00 00 00 00 00 00 00 00 CRC

Verification error or execution Fail return: 0xCB 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 CRC

Reply description: AA BB CC DD is the target step value, 4 bytes, low byte first

9. Read the power of the device:

Command format:	0x13 AA 00 00 00 00 00 00 00 00 00 00 00 00 00 00 CRC
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AA: 0X99, now perform a battery test

Function:	read the power of the device
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describe:

command reply:

The verification is correct and the execution OK returns: 0x13 AA 00 00 00 00 00 00 00 00 00 00 00 00 00 00 CRC

Verification error or execution Fail returns: 0x93 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 CRC

Reply description: AA power level, the value is 0 to 100, indicating the percentage of power.

10 Read the MAC address:

Command format:	0x22 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 CRC
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Function: Read the MAC address of the device.

describe:

command reply:

The verification is correct and the execution OK returns: 0x22 MAC0 MAC1 MAC2 MAC3 MAC4 MAC5 00 00 00 00 00 00 00 00 CRC

Verification error or execution Fail returns: 0xA2 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 CRC

11. Read software version number

API

Command format: 0x27 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 CRC

Function: read software version number.

Command reply:

The verification is correct and the execution OK returns: 0x27 AA BB CC DD EE FF GG 00 00 00 00 00 00 ID
CRC

Verification error or execution Fail return: 0xA7 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 CRC

Response description:

Expressed as

AA BB CC DD is the software version number (in hexadecimal BCD code format)

FF GG HH is the corresponding time

21. Restore factory settings:

Command format: 0x12 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 CRC

Function: restore to the factory minimum current

describe:

command reply:

The verification is correct and the execution OK returns: 0x12 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
CRC

Verification error or execution Fail returns: 0x92 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 CRC

12. MCU soft reset command

Command format: 0x2E 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 CRC

Function: APP reset main control MCU

describe:

command reply:

The verification is correct and the execution OK returns: 0x2E 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
CRC

Verification error or execution Fail return: 0xAE 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 CRC

13. Transmit motor vibration signal

Command format: 0x36 AA 00 00 00 00 00 00 00 00 00 00 00 00 00 00 CRC

Function: set display mode

Description: AA: Vibration times, the unit is 0.5S for each vibration and 0.5 for stop. The maximum value range is 5.

command reply:

The verification is correct and the execution OK returns: 0x36 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
CRC

Verification error or execution Fail returns: 0xB6 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 CRC

16. Set the time period for automatic heart rate detection:

API

Command format: 0x2A AA BB CC DD EE FF GG HH II 00 00 00 00 00 00 CRC

Function: Set the exercise time period.

describe:

AA: work mode, 0: off 1: time section work mode, 2: interval work mode within time section.

BB: The hour part of the time to start measuring (24-hour clock). BCD code format (such as 23 points

AA = 0x23).

CC is the minute part of the start measurement time, in BCD code format (such as 59 minutes BB = 0x59).

DD is the hour part of the end measurement time (24-hour clock). BCD code format (such as 23 points AA = 0x23).

EE is the minute part of the end measurement time, in BCD code format (such as 59 minutes BB = 0x59).

FF is the day of the week enable bit

Bit0 = 0 means not enabled on Sunday, Bit0 = 1 means enabled on Sunday.

Bit1 = 0 means Monday is not enabled, Bit1 = 1 means Monday is enabled.

Bit2 = 0 means Tuesday is not enabled, Bit2 = 1 means Tuesday is enabled.

Bit3 = 0 means Wednesday is not enabled, Bit3 = 1 means Wednesday is enabled.

Bit4 = 0 means Thursday is not enabled, Bit4 = 1 means Thursday is enabled.

Bit5 = 0 means Friday is not enabled, Bit5 = 1 means Friday is enabled.

Bit6 = 0 means Saturday is not enabled, Bit6 = 1 means Saturday is enabled.

GG HH: When working in working mode 2, how much time to test once, the unit is minute. Each test takes 1 minute.

II: 1. Set heart rate 2. Set blood oxygen 3. Set temperature 4. HRV measurement

command reply:

The verification is correct and the execution OK returns: 0x2A 00 00 00 00 00 00 00 00 00 00 00 00 00 00 CRC

Verification error or execution Fail return: 0xAA 00 00 00 00 00 00 00 00 00 00 00 00 00 00 CRC

17. Read the automatic heart rate detection period:

Command format: 0x2B AA 00 00 00 00 00 00 00 00 00 00 00 00 00 CRC

Function: APP reads the time period value set by the device.

Description: AA : 1. Read heart rate setting time 2. Read and set blood oxygen time 3. Read and set temperature time 4. Read and set HRV measurement time

command reply:

The verification is correct and the execution OK returns: 0x2B AA BB CC DD EE FF GG HH II 00 00 00 00 CRC

Verification error or execution Fail returns: 0xAB 00 00 00 00 00 00 00 00 00 00 00 00 00 00 CRC

Response description:

AA: work mode, 0: off 1: time section work mode, 2: interval work mode within time section.

BB: The hour part of the time to start measuring (24-hour clock). BCD code format (such as 23 points

AA = 0x23).

CC is the minute part of the start measurement time, in BCD code format (such as 59 minutes BB = 0x59).

API

DD is the hour part of the end measurement time (24-hour clock). BCD code format (such as 23 points AA = 0x23).

EE is the minute part of the end measurement time, in BCD code format (such as 59 minutes BB = 0x59).

FF is the day of the week enable bit

Bit0 = 0 means it is not enabled on Sunday, Bit0 = 1 means it is enabled on Sunday.

Bit1 = 0 means Monday is not enabled, Bit1 = 1 means Monday is enabled.

Bit2 = 0 means Tuesday is not enabled, Bit2 = 1 means Tuesday is enabled.

Bit3 = 0 means Wednesday is not enabled, Bit3 = 1 means Wednesday is enabled.

Bit4 = 0 means Thursday is not enabled, Bit4 = 1 means Thursday is enabled.

Bit5 = 0 means Friday is not enabled, Bit5 = 1 means Friday is enabled.

Bit6 = 0 means Saturday is not enabled, Bit6 = 1 means Saturday is enabled.

GG HH: When working in working mode 2, how much time to test once, the unit is minute. Each test takes 1 minute.

18. Set the alarm clock (variable length command):

Command format: 0x23 XX AA BB CC DD EE FF GG T1 T2 ... T30

The variable length is to select the command according to the actual maximum length of the mobile phone. For example, the single length of this instruction is 37 bytes, if the maximum length of the mobile phone is 200 bytes. That one command can send $200/37=5.4$, which is 5 after rounding up, that is to say, one command can set 5 alarm clocks.

The instruction instance structure is as follows.

0x23 05 00 BB CC DD EE FF GG T1 T2 ... T30 0x23 05 01 BB CC DD EE FF GG T1 T2 ... T30 0x23 05 02 BB CC DD EE FF GG T1 T2 ... T30 0x23 05 03 BB CC DD EE FF GG T1 T2 ... T30 0x23 05 04 BB CC DD EE FF GG T1 T2 ... T30 **0x23 FF (indicates that it is the end, and the data will be saved at this time)**

Function: Set the alarm clock function.

describe:

XX Total number of alarm clocks used.

The range of AA is 0 to 9 alarm clock numbers, (the value is 0 to 9), and 10 alarm clocks can be set.

BB 0 is Disable; 1 is Enable

CC alarm clock type, 1 alarm clock, 2 reminders to take medicine, 3: reminders to drink water.

DD is the hour part of the alarm time (24-hour system), in BCD code format (such as 23 o'clock CC = 0x23).

EE is the minute part of the alarm time, in BCD code format (such as 59 minutes CC = 0x59).

FF week is enabled. Bit0 = 0 means it is not enabled on Sunday, Bit0 = 1 means it is enabled on Sunday.

Bit1 = 0 means Monday is not enabled, Bit1 = 1 means Monday is enabled.

Bit2 = 0 means Tuesday is not enabled, Bit2 = 1 means Tuesday is enabled.

Bit3 = 0 means Wednesday is not enabled, Bit3 = 1 means Wednesday is enabled.

Bit4 = 0 means Thursday is not enabled, Bit4 = 1 means Thursday is enabled.

Bit5 = 0 means Friday is not enabled, Bit5 = 1 means Friday is enabled.

Bit6 = 0 means Saturday is not enabled, Bit6 = 1 means Saturday is enabled.

GG: Length, 0 to 30, indicates the length of the following text display

T1 T2 ... T30, the content of the text, regardless of the length in front, the useless position is represented by 0, and the content is ASCII coded.

Command reply: 0x23 R1 23 R2 ... 23 RX

IDX: Indicates which alarm clock it is, RX: 0xEE: Indicates that the alarm clock will be saved at this time.

1.9 reminder command

Command format: 0x4D AA BB T1 T2 T3 T60 CC X 1 X 2 X 3 X15

Function: Reminder command. The maximum length of the reminder message is 60bytes.

Description: AA reminder command type

0: Incoming call

1 Mobile phone information, other APPs also belong to information.

2. WeChat, 3, facebook, 4, instagram, 5, skype, 6, telegram, 7, tweetie2, 8, vkclient

9. WhatsApp 10. QQ 11.in

0XFF Stop incoming call reminder

BB length: no more than 60 bytes.

T1 T2 T3 ... T60: Information content.

CC: The length of the contact , the maximum is 15

$X_1 X_2 X_3 \dots X_{15}$: information content

command reply:

The verification is correct and the execution OK returns: 0x4D 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
CRC

Verification error or execution Fail returns: 0xCD 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 CRC

20. Set the exercise period:

Command format: 0x25 AA BB CC DD EE FF GG HH 00 00 00 00 00 00 CRC

Function: Set the exercise time period.

describe:

AA is the hour part (24-hour clock) of the start time of the exercise. BCD code format (such as 23 points AA = 0x23).

BB is the minute part of the start time of exercise, in BCD code format (such as 59 minutes BB = 0x59).

CC is the hour part of the end exercise time (24-hour clock). BCD code format (such as 23 points AA = 0x23).

DD is the minute part of the end exercise time, in BCD code format (such as 59 minutes BB = 0x59).

EE is the day of the week enable bit

Bit0 = 0 means not enabled on Sunday, Bit0 = 1 means enabled on Sunday.

Bit1 = 0 means not enabled on Monday, Bit1 = 1 means enabled on Monday.

Bit2 = 0 means not enabled on Tuesday, Bit2 = 1 means enabled on Tuesday.

Bit3 = 0 means not enabled on Wednesday, Bit3 = 1 means enabled on Wednesday.

Bit4 = 0 means not enabled on Thursday, Bit4 = 1 means enabled on Thursday.

Bit5 = 0 means not enabled on Friday, Bit5 = 1 means enabled on Friday.

Bit6 = 0 means Saturday is not enabled, Bit6 = 1 means Saturday is enabled.

FF exercise reminder period, the unit is minute, the value is (1-255). For example, 255 means a reminder every 255 minutes.

API

GG minimum number of steps

HH enable switch, 1: enable, 0: disable

command reply:

The verification is correct and the execution OK returns: 0x25 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 CRC

Verification error or execution Fail return: 0xA5 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 CRC

21. Read exercise period:

Command format: 0x26 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 CRC

Function: APP reads the exercise time period value set by the device.

describe:

command reply:

The verification is correct and the execution OK returns: 0x26 AA BB CC DD EE FF 00 00 00 00 00 00 00 00 CRC

Verification error or execution Fail return: 0xA6 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 CRC

Response description:

AA is the hour part (24-hour clock) of the start time of the exercise. BCD code format (such as 23 points AA = 0x23).

BB is the minute part of the start time of exercise, in BCD code format (such as 59 minutes BB = 0x59).

CC is the hour portion (24-hour clock) of the end exercise time. BCD code format (such as 23 points AA = 0x23).

DD is the minute part of the end exercise time, in BCD code format (such as 59 minutes BB = 0x59).

EE is the week enable bit

Bit0 = 0 means it is not enabled on Sunday, Bit0 = 1 means it is enabled on Sunday.

Bit1 = 0 means Monday is not enabled, Bit1 = 1 means Monday is enabled.

Bit2 = 0 means Tuesday is not enabled, Bit2 = 1 means Tuesday is enabled.

Bit3 = 0 means Wednesday is not enabled, Bit3 = 1 means Wednesday is enabled.

Bit4 = 0 means Thursday is not enabled, Bit4 = 1 means Thursday is enabled.

Bit5 = 0 means Friday is not enabled, Bit5 = 1 means Friday is enabled.

Bit6 = 0 means Saturday is not enabled, Bit6 = 1 means Saturday is enabled.

FF exercise reminder period, the unit is minute, the value is (1-255). For example, 255 means a reminder every 255 minutes.

HH enable switch, 1: enable, 0: disable

22. Obtain the total step count data

Command format: 0x51 AA 00 00 00 00 00 00 00 00 00 00 00 00 00 00 CRC

AA: 99: delete historical total data, 0: read total data.

Function: Get all total data.

return data. (Variable length instruction) 0x51 ID YY MM DD S1 S2 S3 S4 T1 T2 T3 T4 D1 D2 D3 D4 K1 K2 K3 K4 DA DB F1 F2 F3 F4

The minimum unit is 27 bytes. How many days of total data a BLE data packet contains can be dynamically controlled according to the length of the actual maximum sent packet.

ID: number, the total data of the last X days. 0 means the current day, which is real-time data. The range is 0 to 29. A total of 30 days of data.

API

YY MM DD: Indicates the date of the data: year month day
S1 S2 S3 S4: Number of steps: high order follows.
T1 T2 T3 T4: Movement time: the high bit comes after, the unit is second.
D1 D2 D3 D4: distance, the high position is behind, the unit is 0.01KM
K1 K2 K3 K4: Calories, the high order is the last, the unit is 0.01KCAL
DA DB : target: the range is 0 to 65535, the high bit comes after
F1 F2 F3 F4: Fast motion time, the high bit is last, the unit is minute.

23. Obtain detailed data of steps

Command format: 0x52 AA BB CC 00 00 00 00 00 00 00 00 00 00 00 00 CRC

AA: 99: Delete detailed step data,

- 0: Read the latest detailed step data.
- 1: Read the step number detailed data at the specified position.
- 2: Continue to read the next piece of data at the last position.

Function: Obtain detailed data of steps.

return data. (Variable length instruction) 0x52 ID1 ID2 YY MM DD HH mm SS S1 S2 K1 K2 D1 D2 SD1 SD2 SD3 ... SD10

The minimum unit is 25 bytes. How many days of total data a BLE data packet contains can be dynamically controlled according to the length of the actual maximum sent packet.

ID1 ID2: Read data number, the high bit follows. .
YY MM DD HH mm SS: Indicates the date of the data: year, month, day, hour, minute, second
S1 S2: Number of steps: high position last.
K1 K2: Calories, the high digit comes after, the unit is 0.01KCAL
D1 D2: Distance, with the high position at the end, the unit is 0.01KM
SD1 SD2 SD3 ... SD10: Indicates the number of steps per minute within the 10 minutes.

24. Get detailed sleep data

Command format: 0x53 AA BB CC 00 00 00 00 00 00 00 00 00 00 00 00 CRC

AA: 99: Delete sleep details,

- 0: Read recent sleep detailed data.
- 1: Read the sleep detailed data at the specified location.
- 2: Continue to the next piece of data at the last read position.

return data. (variable length instruction) 0x53 ID1 ID2 YY MM DD HH mm SS LEN SD1 SD2 SD3 ... SD24

The minimum unit is 34 bytes. How many days of total data a BLE data packet contains can be dynamically controlled according to the length of the actual maximum sent packet.

ID1 ID2: Read data number, the high bit follows. .
YY MM DD HH mm SS: Indicates the date of the data: year, month, day, hour, minute, second
LEN: The length indicates the length of the sleep data.

API

SD1 SD2 SD3 ... SD24: Each number represents the sleep quality of 5 minutes, and it represents the sleep data of 2 hours at most.

25. Get heart rate data

Command format: 0x54 AA BB CC 00 00 00 00 00 00 00 00 00 00 00 00 CRC

AA: 99: delete all heart rate data,

- 0: Read the latest heart rate data.
- 1: Read the heart rate data at the specified location.
- 2: Continue to read the next piece of data at the last position.

return data. (variable length instruction) 0x54 ID1 ID2 YY MM DD HH mm SS SD1 SD2 SD3 ... SD12

The minimum unit is 21 bytes. How many days of total data a BLE data packet contains can be dynamically controlled according to the length of the actual maximum sent packet.

ID1 ID2: Read data number, the high bit follows. .

YY MM DD HH mm SS: Indicates the date of the data: year, month, day, hour, minute, second

SD1 SD2 SD3 ... SD12: Save a heart rate value every 10 seconds, a total of 12 heart rate values, representing 2 minutes of heart rate data.

26. Obtain a single heart rate data (interval heart rate test)

Command format: 0x55 AA BB CC 00 00 00 00 00 00 00 00 00 00 00 00 CRC

AA: 99: delete all single heart rate data,

- 0: Read the latest single heart rate data.
- 1: Read a single heart rate data at the specified location.
- 2: Continue to read the next piece of data at the last position.

return data. (variable length instruction) 0x55 ID1 ID2 YY MM DD HH mm SS HH

The smallest unit is 10 bytes. How many days of total data a BLE data packet contains can be dynamically controlled according to the length of the actual maximum sent packet.

ID1 ID2: Read data number, the high bit follows. .

YY MM DD HH mm SS: Indicates the date of the data: year, month, day, hour, minute, second

HH: heart rate value.

28. Get alarm clock data

Command format: 0x57 AA BB CC 00 00 00 00 00 00 00 00 00 00 00 00 CRC

AA: 99: delete alarm clock data,

- 0: Read alarm clock data.

return data. (variable length instruction) 0x57 ID1 ID2 AA BB CC DD EE FF T1 T2 ... T30

describe:

ID1 ID2: Read data number, the high bit follows.

The range of AA is 0 to 4 alarm clock numbers, (the value is 0 to 4), and 5 alarm clocks can be set.

API

BB alarm clock type, 0: Disable; 1: Enable alarm clock, 2: Reminder of taking medicine, 3: Reminder of drinking water.

CC is the hour part of the alarm time (24-hour system), in BCD code format (such as 23 o'clock CC = 0x23).

DD is the minute part of the alarm time, in BCD code format (such as 59 minutes CC = 0x59).

EE week is enabled. Bit0 = 0 means it is not enabled on Sunday, Bit0 = 1 means it is enabled on Sunday.

Bit1 = 0 means Monday is not enabled, Bit1 = 1 means Monday is enabled.

Bit2 = 0 means Tuesday is not enabled, Bit2 = 1 means Tuesday is enabled.

Bit3 = 0 means Wednesday is not enabled, Bit3 = 1 means Wednesday is enabled.

Bit4 = 0 means Thursday is not enabled, Bit4 = 1 means Thursday is enabled.

Bit5 = 0 means Friday is not enabled, Bit5 = 1 means Friday is enabled.

Bit6 = 0 means Saturday is not enabled, Bit6 = 1 means Saturday is enabled.

FF: Length, 0 to 30, indicates the length of the following text display

T1 T2 ... T30, the content of the text, regardless of the length in front, the useless position is represented by 0, and the content is ASCII coded.

28. Get sleep debug data

Command format: 0x58 AA BB CC 00 00 00 00 00 00 00 00 00 00 00 00 CRC

AA: 99: sleep debug data,

1: Read the sleep debug data at the specified location.

2: Continue to read the next piece of data at the last position.

Return data. (variable length instruction) 0x58 ID1 ID2 AA BB CC DD EE FF T1 T2 ... T

Description: This computer APP reads data and does not need to use this interpretation.

29. Combination package sending and receiving:

Command format: 0x99 LEN A1 A2 ... A16 B1 B2 ... B16 C1 C2 ... C16 X1 X2 ... X16.

Function: Combination packet sending and receiving.

describe:

LEN: length, indicating how long the subsequent data is. Must be an integer multiple of 16.

A1 A2 ... A16: The first instruction.

B1 B2 ... B16: The second instruction.

C1 C2 ... C16: The third instruction.

...

X1 X2 ... X16: The Xth instruction.

28. Obtain sports data (here refers to the data in sports mode, such as running, cycling...)

API

Command format: 0x5C AA BB CC 00 00 00 00 00 00 00 00 00 00 00 00 CRC

AA: 99: delete all exercise data,

1: Read the motion data of the specified position.

2: Continue to read the next piece of data at the last position.

Return data. (Variable length instruction) 0x5C ID1 ID2 YY MM DD HH mm SS TY HH tt tt SS SS PP PP KK
KK KK KK DD DD DD DD

YY MM DD HH mm SS: time: year, month, day, hour, minute, second

TY: sports type

0: Run,

1: Cycling

7.Breath

HH : heart rate

Tt tt: exercise time

SS SS: number of motion steps

PP PP: movement speed

KK KK KK KK: Calories

DD DD DD DD: distance.

27. Get HRV test data

Command format: 0x56 AA BB CC 00 00 00 00 00 00 00 00 00 00 00 00 CRC

AA: 99: delete HRV data,

0: Read the latest HRV data.

1: Read the HRV data at the specified location.

2: Continue to read the next piece of data at the last position.

return data. (variable length instruction) 0x56 ID1 ID2 YY MM DD HH mm SS D1 D2 D3 D4 00 00

The minimum unit is 15 bytes. How many days of total data a BLE data packet contains can be dynamically controlled according to the length of the actual maximum sent packet.

ID1 ID2: Read data number, the high bit follows. .

YY MM DD HH mm SS: Indicates the date of the data: year, month, day, hour, minute, second

D1: HRV value.

D2: empty

D3: empty

D4: degree of fatigue.

00 00: reserved

API

30. Sport mode control switch

Command format: 0x1 9 AA BB CC DD 00 00 00 00 00 00 00 00 00 00 CRC

AA: 1: start, 4 end.

BB: Sport mode type.

0: Run,

1: Cycling

6. Yoga

return data. 0x19 AA 00 00 00 00 00 00 00 00 00 00 00 00 00 00 CRC

AA: 1: Success

0: Failed, if the wristband is testing blood pressure or is already in the exercise mode and restarts the exercise through the APP, it will return failure.

Verification error return: 0xA6 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 CRC

31 APP heartbeat data packets.

When the exercise mode of the bracelet is turned on through the APP, a heartbeat data packet will be sent to the bracelet every second.

Command format: 0x17 D1 D2 D3 D4 MM SS GG 00 00 00 00 00 00 00 00 CRC

D1 D2 D3 D4: distance, floating point number. The unit is KM.

MM SS: distribution speed, MM: minute, SS second. The minimum pace is 255 minutes and 59 seconds.

GG: Cell phone GPS signal strength. The range is: 1, 2, 3

3 2 Wristband heartbeat data packets.

When the exercise mode of the bracelet is turned on through the APP, a heartbeat data packet will be sent to the mobile phone every second.

Command format: 0x1 8 HR S1 S2 S3 S4 K1 K2 K3 K4 T1 T2 T3 T4 00 00 00 00 CRC

HR: 0xFF: Indicates that the bracelet ends exercise mode, other values are heart rate values,

S1 S2 S3 S4: Step count.

K1 K2 K3 K4: Calories, float.

T1 T2 T3 T4: Exercise time

37. Obtain blood oxygen data (manual test)

Command format: 0x 60 AA BB CC 00 00 00 00 00 00 00 00 00 00 00 00 CRC

AA: 99: delete all single blood oxygen data,

0: Read the latest single blood oxygen data.

1: Read the single blood oxygen data at the specified location.

2: Continue to read the next piece of data at the last position.

return data. (variable length instruction) 0x 60 ID1 ID2 YY MM DD HH mm SS HH

API

The smallest unit is 10 bytes. How many days of total data a BLE data packet contains can be dynamically controlled according to the length of the actual maximum sent packet.

ID1 ID2: Read data number, the high bit follows.

YY MM DD HH mm SS: represents the date of the data: year, month, day, hour, minute and second.

HH: blood oxygen level.

38. Obtain blood oxygen data (automatic test)

Command format: 0x 66 AA BB CC 00 00 00 00 00 00 00 00 00 00 CRC

AA: 99: delete all single blood oxygen data,

0: Read the latest single blood oxygen data.

1: Read the single blood oxygen data at the specified location.

2: Continue to read the next piece of data at the last position.

Return data. (Variable length instruction) 0x 66 ID1 ID2 YY MM DD HH mm SS HH

The smallest unit is 10 bytes. How many days of total data a BLE data packet contains can be dynamically controlled based on the actual maximum length of the sent packet.

ID1 ID2: Read data number, the high bit follows.

YY MM DD HH mm SS: Indicates the date of the data: year, month, day, hour, minute, second

HH: blood oxygen level.

39. Obtain temperature data (the temperature measured by the temperature test interface).

Command format: 0x62 AA BB CC 00 00 00 00 00 00 00 00 00 00 CRC

AA: 99: delete all exercise data,

1: Read the motion data of the specified position.

2: Continue to read the next piece of data at the last position.

return data. (variable length instruction) 0x62 ID1 ID2 YY MM DD HH mm SS T 1 T2

YY MM DD HH mm SS: time: year, month, day, hour, minute, second

T1 T2 : Temperature value, the high digit comes after, and 1 decimal place is reserved.

40. Obtain temperature data (problems detected by automatic test).

Command format: 0x6 5 AA BB CC 00 00 00 00 00 00 00 00 00 00 CRC

AA: 99: delete all exercise data,

1: Read the motion data of the specified position.

2: Continue to read the next piece of data at the last position.

return data. (variable length instruction) 0x6 5 ID1 ID2 YY MM DD HH mm SS T 1 T2

YY MM DD HH mm SS: time: year, month, day, hour, minute, second

T1 T2 : Temperature value, the high digit comes after, and 1 decimal place is reserved.

33 health measurement control

Command format: 0x 28 AAB B CC 00 00 00 00 00 00 00 00 00 00 CRC

Function: APP initiates heart rate , blood oxygen, and blood pressure measurements to the device

API

AA:

0x01, H RV measurement

BB:

1 Turn on measurement

0 close measurement

0x02, heart rate measurement

BB:

1 Start measurement

0 close measurement

0x03, blood oxygen measurement

BB:

1 Start measurement

0 Close measurement

0x04, temperature measurement

BB:

1 Start measurement

0 Close measurement

command reply:

The verification is correct and the execution OK returns every second: 0x 28 AABB CC DD EE
FF GG HH II 00 00 00 00 00 00 CRC

AA: the current measurement type, as described above

BB: heart rate value

CC: blood oxygen value (you can ignore it if it is not in blood oxygen mode)

DD: HRV value

EE: fatigue level

FF: high blood pressure

GG: blood pressure low pressure

HH II: temperature value

34. MCU shutdown command

Command format: 0x 12 0 1 00 00 00 00 00 00 00 00 00 00 00 00 00 00 CRC

Function: APP shuts down the main control MCU

describe:

command reply:

The verification is correct and the execution is OK. Return: 0x 12 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 CRC

Verification error or execution Fail return: 0x ca 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 CRC